A069 v0.6.3 — FULL GPU

Intel Arc A770 / checkpoint continuation / source-native high-resolution 3D GP
Development engineering report · 19 October 2026 · NOT FROZEN / NOT PHYSICS CERTIFIED

Scope and runtime

Implements an actual full 3D split-step Strang GP evolution runner using PyTorch-XPU complex64 on the Intel Arc A770; reuses unchanged source-native v0.6.2 physics, explicit mu=1/(2 xi²), and hard no-fallback hardware selection. Two supplied Windows paths are registered as read-only B160 and A160 checkpoints, both at absolute step 16. A newly initialized B192/B256 or A512 run is built independently from pinned 2015 geometry, not interpolated from a coarse evolved checkpoint.

Case / role	Grid; xi; mu	Initial state	Geometry G32
B160 — primary	160³; 0.28; 6.377551	Existing B160 T000016.npz	PASS screen only
A160 — diagnostic	160³; 0.08; 78.125	Existing A160 T000016.npz	FAIL; explicit override
B192 / B256	192³ / 256³; 0.28; 6.377551	Fresh original 2015 source	PASS screen only
A512 — conditional	512³; 0.08; 78.125	Fresh original 2015 source	PASS screen only

Hardware evidence and precision

Earlier user-provided Arc A770 tests demonstrated XPU complex FFT and GP numerical screening on synthetic N32/N64 and a genuine B160 checkpoint. The B160 preflight recorded 4-step relative L2 versus NumPy FP64 of 4.11e-7, GPU time 0.13305 seconds versus CPU 2.19727 seconds, and approximately 290 MiB peak reserved PyTorch memory. These results are from a previous user-executed GPU preflight, not from this new FULL runner. They do not certify time-resolved trefoil topology, float64 arithmetic on the Arc, or FFT workspace peak at N512.

Locally executed validation of this release

56/56 Python regressions PASS (inherited v0.6.2 suites and new v0.6.3 engine tests). A full CPU-only engineering smoke executed actual source-native B48 GP from absolute step 0 to 2, saved hash-bound complex64 checkpoints, then resumed from step 2 to 4; 2-step NumPy FP64 parity PASS on each segment. Separate offline subgrid topology analysis executed and returned G33 PASS as DIAGNOSTIC_ONLY. Actual Arc XPU executions of this new v0.6.3 runner, and B192/B256/A512 campaigns, are NOT EXECUTED HERE.

Important FP32 topology audit repair

The inherited tracker enforces noninteger plaquette winding residual below 3e-11 and can correctly fail if link phases are computed in float32. The GPU saves complex64 fields. The offline analyzer converts each saved field to complex128 before phase-link winding, without weakening the strict winding tolerance. This avoids confusing float32 rounding with a physical lattice discontinuity. Any diagnostic is still not an isotopy certificate.

Execution and scientific qualification

Windows commands

After extracting the ZIP, reuse the installed .venv-xpu Python that passed the prior GPU preflight. Set the environment variable A069_XPU_PYTHON if this Python is located in a different folder. Then run:

run_known_checkpoints.cmd — checks the exact user-provided B160 and A160 paths; continues each from step 16 to step 32; saves new immutable checkpoints and JSON evidence.

run_optional_b256.cmd — fresh original-source Torus B256 initialization in RAM-bounded slabs; performs 16 XPU GP steps; generates new checkpoint and diagnostics. run_optional_a512.cmd requires confirmation and the strict VRAM guard may reject the job. This is intentional.

Post-process output with analyze_snapshot.py --field results\B160\RUN_ID\T000032.npy --out .... Multiple snapshots are needed for event tracking; two equal determinant-3 projections are insufficient for G34.

VRAM and numerical safety

Conservative screening guard checks physical free VRAM on the selected Arc and initially budgets 112 bytes/voxel plus 0.75 GiB of reserve at 88% utilization. An optional measured B256 full XPU result may calibrate bytes/voxel with a positive margin; the guard never becomes a free overcommit switch. Native FFT workspace may not be visible to PyTorch peak-reserved counters. The CPU original-source initializer works in thin slabs and itself can remain time-consuming for N512. On-device Hamiltonian uses FFT Parseval instead of three extra inverse gradient FFTs; energy comparisons need independent CPU-FP64 confirmation.

Framework v1.0.10: actual GateLedger adapter vs certification

The package includes a conservative JSON evidence export and a second adapter that imports and invokes the actual installed framework v1.0.10 GateLedger.run_evaluators, with exact version/path verification. This adapter was smoke-tested locally using the v1.0.10 ledger source; it is not an officially signed BLIND framework run, released frozen framework artifact, or physical G9 reveal. The Drive release consulted identifies v1.0.10 as LOCAL_VALIDATED_CANDIDATE; native full solver qualification remains separate.

Gate	Outcome/authority	Required next validation
G32: geometry / mesh	A160 FAIL; B160/B256 and A512 PASS screen only	Do not call geometry readiness 3D stability
G33: filament projection	Offline snapshot diagnostics; optional PASS DIAGNOSTIC_ONLY	Dense snapshots; subcell shifts; ambiguity tests
G34: knot topology	UNRESOLVED	Time-resolved ambient isotopy, mesh/time convergence, event audit
G35: knot internal clock	UNRESOLVED	Independent stable RPO/Floquet eigenmode, >=4 cycles, blind holdout

Deliverables / integrity

The release ZIP includes self-contained unmodified v0.6.2 analytical/tracking modules, pinned public v0.5.0 solver archive, source-native GPU runner, checkpoint reader/writer, offline analyzer, two framework adapters, Windows .cmd launchers, regression tests, source SHA-256 manifest and development report. No private blind key material is included. Historical v0.5.0/v0.6.2 outputs remain immutable.

Epistemic restriction: GP is a compressible order-parameter model, not the SST incompressible Euler substrate. Neither an independently validated long-lived vortex knot nor a frame-/gauge-independent Swirl Clock eigenfrequency is established. G34 and G35 remain UNRESOLVED.